Projected gradient descent

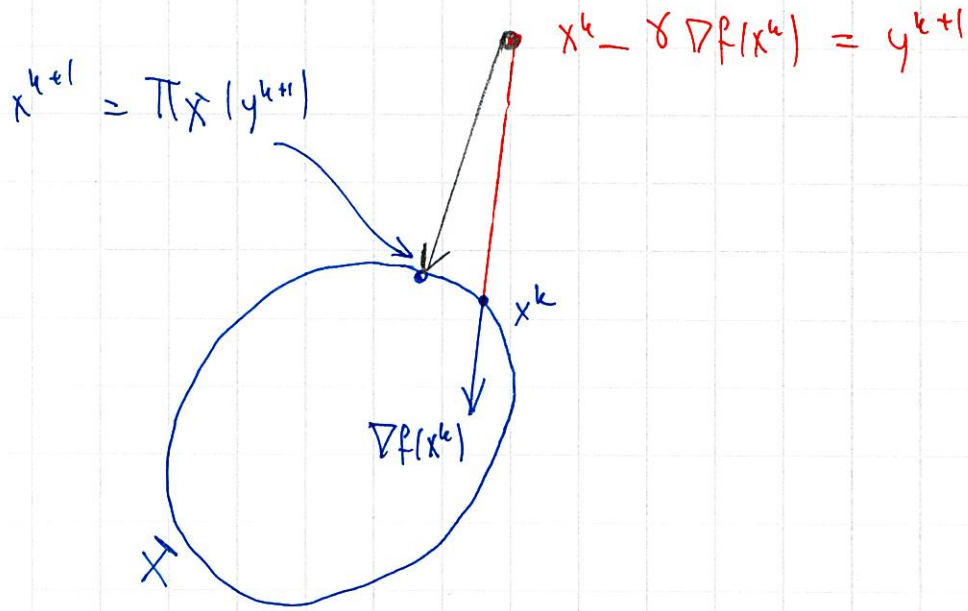
- For convex function $f: \mathbb{R}^n \to \mathbb{R}$ and convex set $X \subseteq \mathbb{R}^n$, consider the constrained optimization problem

$$\min_{x \in X} f(x)$$

- Projected gradient descent works as follows. Start at some initial point x^0, generate a sequence of points $y^1, x^1, y^2, x^2, y^3, x^3, \ldots$ as

$$y^{k+1} = x^k - \gamma \nabla f(x^k)$$

$$x^{k+1} = \Pi_X(y^{k+1})$$

- 

- Definition of projection:

$$\Pi_X(y) = \arg\min_{x \in X} \|x-y\|^2$$

- Let's show an important property of projections.

Recall: if g is convex, $g(w) = g(z) + \nabla^t g(z)(w-z) + o(\|w-z\|)$ (Taylors)

Recall: $x^t y = \|x\| . \|y\| . \cos\theta$, where θ is the angle between x and y.

Thus, if $x^t y \leq 0$, then the angle between x and y is obtuse

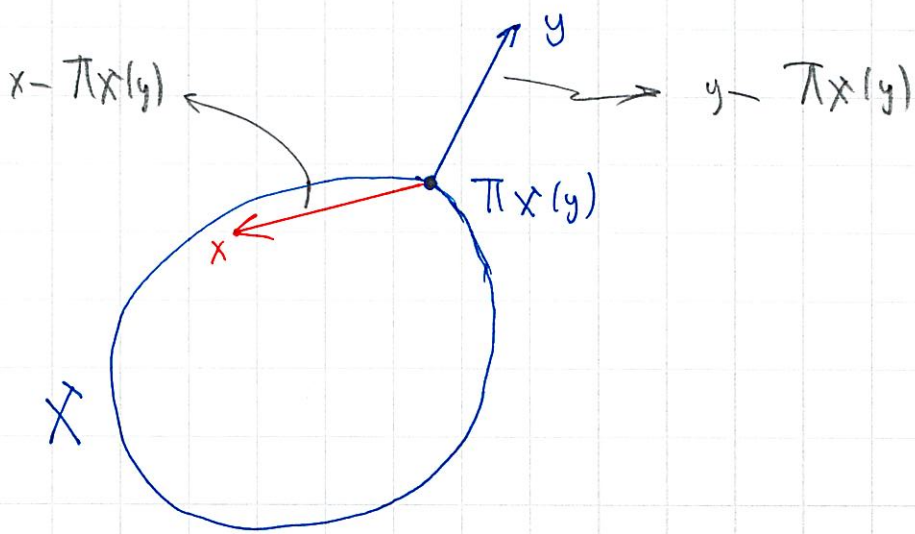
- Note let $g(x) = \|x\|^2$, compute $\nabla g(x)$

$$g(x) = \sum_{i=1}^{n} (x_i)^2 \qquad \frac{\partial g(x)}{\partial x_i} = 2 \cdot x_i$$

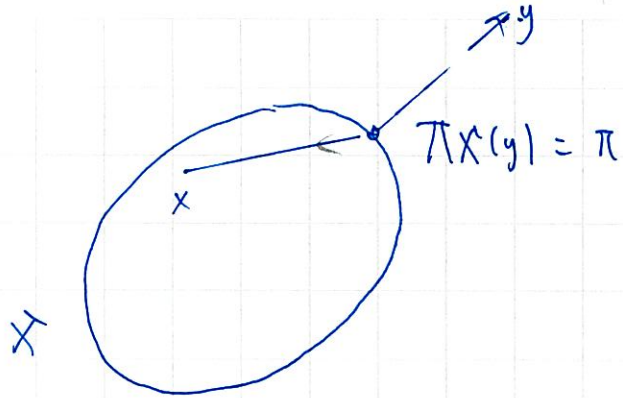
$$\nabla g(x) = \left(\frac{\partial g(x)}{\partial x_1}, \dots, \frac{\partial g(x)}{\partial x_n} \right) = (2x_1, \dots, 2x_n) = 2 \cdot x$$

- Important property of projections: let X be a convex set, $x \in X$ and $y \in \mathbb{R}^n$. We have

$$(x - \Pi_X(y))^t (y - \Pi_X(y)) \leq 0$$



Pf:



we want to show: $(x-\pi)^t (y-\pi) \leq 0$

- For any $\lambda \in [0,1]$, because X is convex, $x \in X$, $\pi \in X$, we have
$$\lambda x + (1-\lambda)\pi \in X$$

- $\pi = \underset{z \in X}{\arg\min} \|z-y\|^2$. Because $\lambda x + (1-\lambda)\pi \in X$, we must have

$$\|y-\pi\|^2 \leq \|y-(\lambda x + (1-\lambda)\pi)\|^2 = \|y-\pi+\lambda(\pi-x)\|^2 \qquad (\ddagger)$$

- Let's apply Taylor's theorem w/ $y-\pi+\lambda(\pi-x)$ and $z = y-\pi$ and $g(x) = \|x$

$$g(y-\pi+\lambda(\pi-x)) = g(y-\pi) + \nabla^t g(y-\pi)\cdot(\cancel{y-\pi}+\lambda(\pi-x)-\cancel{(y-\pi)}) + o(\|\cancel{y-\pi}+\lambda(\pi-x)-\cancel{(y-\pi)}\|)$$

$$\underbrace{\|y-\pi+\lambda(\pi-x)\|^2}_{\geq \|y-\pi\|^2 \text{ by } (\ddagger)} = \|y-\pi\|^2 + 2\lambda(y-\pi)^t(\pi-x) + o(\lambda\|\pi-x\|)$$

- $\cancel{\|y-\pi\|^2} \leq \cancel{\|y-\pi\|^2} + 2\lambda\,(y-\pi)^t\,(\pi-x) + o(\lambda\,\|\pi-x\|)$ for any $\lambda \in [0,$

$$\Rightarrow \quad 0 \leq 2\lambda\,(y-\pi)^t.(\pi-x) + o(\lambda\,\|\pi-x\|)$$

$$\Rightarrow \quad 0 \leq 2\,(y-\pi)^t\,(\pi-x) + \frac{o(\lambda\,\|\pi-x\|)}{\lambda}$$

$$\Rightarrow \quad 0 \leq 2\,(y-\pi)^t\,(\pi-x) + \frac{o(\lambda\,\|\pi-x\|)}{\lambda\,\|\pi-x\|}\,.\,\|\pi-x\|$$

$$\Rightarrow \quad 0 \leq 2\,(y-\pi)^t\,(\pi-x) + \lim_{\lambda\to 0} \cancel{\frac{o(\lambda\,\|\pi-x\|)}{\lambda.\,\|\pi-x\|}}^{\,0}\,.\,\|\pi-x\|$$

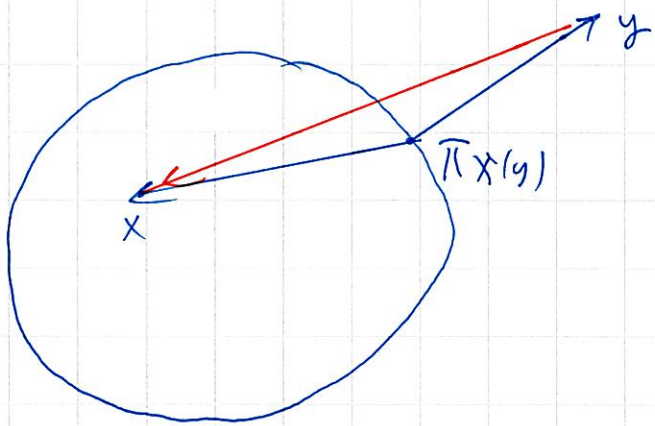
$$\Rightarrow \quad 0 \leq \cancel{2}\,(y-\pi)^t\,(\pi-x) \quad \Rightarrow \quad (y-\pi)^t.(x-\pi) \leq 0$$

done!

- Here's an immediate implication of property of projections:

Let X be convex, $x \in X$ and $y \in \mathbb{R}^n$. Then,

$$\|x-y\|^2 \geq \|x - \Pi_X(y)\|^2 + \|y - \Pi_X(y)\|^2$$



Recall:

$$\|a+b\|^2 = \|a\|^2 + \|b\|^2 + 2\, a^t b$$

Pf: $\pi = \Pi_X(y)$

$$\|x-y\|^2 = \|x - \pi + \pi - y\|^2 = \|x-\pi\|^2 + \|\pi - y\|^2 + 2.(x-\pi)^t(\pi - y)$$

$$= \|x-\pi\|^2 + \|y-\pi\|^2 - 2\underbrace{(x-\pi)^t(y-\pi)}_{\leq 0} \geq \|x-\pi\|^2 + \|y-\pi\|^2$$

from previous result

done!

Convergence of Projected Gradient Descent with Bounded Gradients

- Projected gradient descent is

$$y^{k+1} = x^k - \gamma \cdot \nabla f(x^k) \quad , \quad x^{k+1} = \Pi_X(y^{k+1})$$

- Let's refine the preliminary analysis we did when we proved convergence for gradient descent without projections.

- Let $x^* = \underset{x \in X}{\arg\min}\, f(x)$, we want to bound $f(x^k) - f(x^*)$

By first order characterization of convexity $f(x^*) \geq f(x^k) + \nabla^t f(x^k)(x^* - x^k) \Rightarrow$

$$f(x^k) - f(x^*) \leq \nabla f^t(x^k)(x^k - x^*)$$

- So it is enough to upper bound $\nabla^t f(x^k)(x^k - x^*)$

- $\nabla^t f(x^k)(x^k - x^*) = \frac{1}{\gamma} \cdot (x^k - y^{k+1})^t \cdot (x^k - x^*)$

Recall: $\|a-b\|^2 = \|a\|^2 + \|b\|^2 - 2a^t b \Rightarrow 2a^t b = \|a\|^2 + \|b\|^2 - \|a-b\|^2$

Thus, $\nabla^{+} f(x^k)\,(x^k - x^*) = \frac{1}{2\gamma}\left(\|x^k - y^{k+1}\|^2 + \|x^k - x^*\|^2 - \|(\cancel{x^k} - y^{k+1}) - (\cancel{x^k} - x^*)\|^2 \right)$,

$$= \frac{1}{2\gamma}\left(\|x^k - y^{k+1}\|^2 + \|x^k - x^*\|^2 - \|y^{k+1} - x^*\|^2 \right)$$